A Time Series Analysis of Non-Communicable Diseases in Adults at Queen Elizabeth Central Hospital, Blantyre (2010-2022)

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PRESENTATION OUTLINE

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1.0 INTRODUCTION

- Non-Communicable Diseases (NCDs) are diseases that are not caused by acute infection, and result in long term health treatment.(Or et al., 2018)
- They caused 68% of all deaths (38 million) up from 60% in 2000.(Organization et al., 2018)
- NCDs disproportionately affect people in low and middle-income countries, where more than three quarters of global NCDs deaths (31.4 million) occur (Roth, 2018).
- Africa CDC reported that premature deaths due to NCDs increased by 67% between 1990 and 2017 in Sub-Saharan Africa.
- NCDs accounted for 25.4% burden of diseases in Malawi, with 29% NCDs mortality rate in 2015. (Cohen et al., 2017)

2.0 PROBLEM STATEMENT

- A study by Allain (Allain et al., 2017), shows that 29.9% of patients were diagnosed with NCDs (3052 out of 10191 admissions from 1st January 2013 to 31st December 2014) and 55.7% were adults at Queen Elizabeth Central Hospital. 23.1% of deaths were due to NCDs, and in adults, mortality rate was 55.7%.
- However, there was a need for comprehensive time series analysis to elucidate the dynamic nature of NCDs and drivers of their trends and patterns over time.
- The absence of such a study can lead to challenges in identifying and understanding the temporal patterns and trends in NCD prevalence over time resulting in a lack of insight into the changing dynamics of NCDs.

- In addition to that, it can help in identifying emerging trends, outbreaks, or sudden changes in NCD prevalence
- Without timely detection and intervention, the burden of NCDs may escalate, leading to increased morbidity, mortality, and healthcare costs.
- Therefore, this study did a periodic assessment of the trends and patterns of NCDs and project NCDs in so doing , the study would help to keep track of changes in the prevalence of NCDs and to identify areas of priority research and intervention.
- This study aimed at conducting a time series analysis of NCDs, focusing on all NCDs recorded at Queen Elizabeth Central Hospital in Blantyre, Malawi from January 2010 to December 2022.
- The analysis investigated temporal patterns, trends, and seasonality in NCDs prevalence and project the prevalence of NCDs using historical data.

3.0 AIM OF STUDY

Main Objective and Specific Objectives & Research Hypotheses

- To conduct a time series analysis of the NCDs in adults at Queen Elizabeth Central Hospital in Blantyre,Malawi.

1. To determine the trend of NCDs cases at QECH.
2. To examine the seasonality of NCDs cases at QECH.
3. To forecast NCDs cases at QECH for the year 2025.

1. H_0:There is no significant trend in NCDs cases at QECH from January 2010 to December 2022
2. H_0:There is no seasonal variation in NCDs cases at QECH from January 2010 to December 2022
3. H_0:There is no significant difference in the forecasted and actual values in NCDs cases at QECH

3.3 JUSTIFICATION AND SIGNIFICANCE OF THE STUDY

- NCDs are the leading cause of death globally (Organization et al., 2018), but there is limited studies in Malawi of trends, seasonal patterns and forecasts of NCDs.
- Therefore, this study, assisted in planning future services and monitoring the prevalence of NCDs.
- It is important to identify possible seasonal patterns,trends and forecast prevalence of NCDs at QECH in order to help medical officials at QECH planning and monitoring NCDs

4.0 ETHICS & STUDY LIMITATIONS

- A letter asking for permission to conduct this study from The Catholic University of Malawi Research Ethics Committee was obtained and presented to Queen Elizabeth Central Hospital for approval to collect the data.
- Since a time series analysis depends on time lags, missing data points can affect the data analysis.

5.0 LITERATURE REVIEW

5.1 Theoretical Review

- Social determinants of health (SDH) are the conditions in which people are born, grow, live, work, and age, and the wider set of forces and systems shaping the conditions of daily life (Organization et al., 2018).
- These determinants include factors such as socioeconomic status, education, employment, access to healthcare and physical environments.
- In the context of NCDs, social determinants play a significant role in influencing health outcomes, risk factors, and disparities.
- Socioeconomic factors such as income, education, and occupation influence access to healthcare, healthy food options, safe housing all of which contribute to NCD risk.

5.2 Conceptual Framework

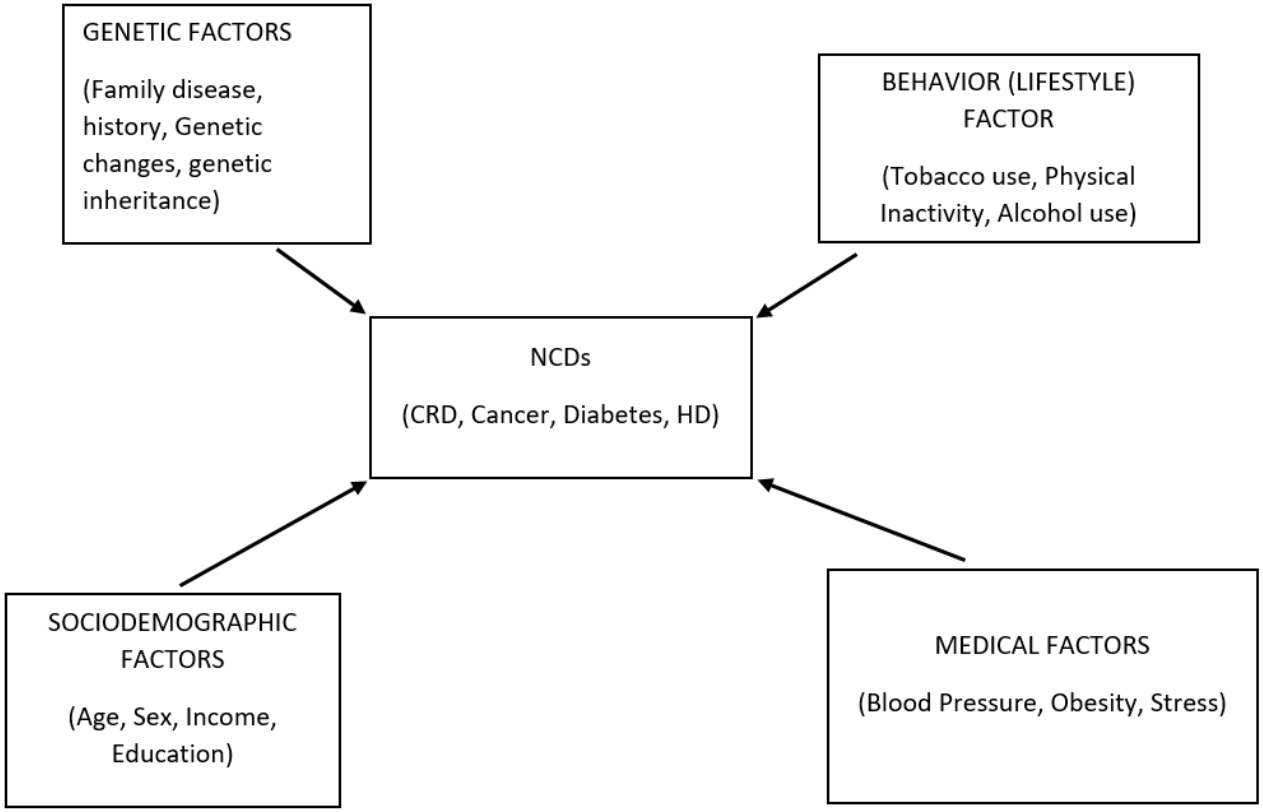

Figure 1: Conceptual Framework

5.3 Empirical Review

- The WHO projects an increasing trend in NCDs and that NCDs will increase by 80% by the year 2030 globally and by 75.6% by the year 2025 in Africa.(Organization et al., 2018)
- According to Mudie (Mudie et al., 2019),in sub-Saharan Africa (SSA), the significance of NCDs is increasing and it is projected that they will overtake infectious diseases as major sources of morbidity and mortality by 2030.
- According to Olds,(Olds et al., 2023), there is an increasing trend in NCDs, a study done in Neno District Hospital has shown.

6.0 NATERIALS & METHODS

6.1 Study Area

- The study was conducted in Blantyre District, Southern Malawi.QECH is the largest hospital in Southern Malawi .It is along Masauko Chipembere Highway and opposite KIA motors.

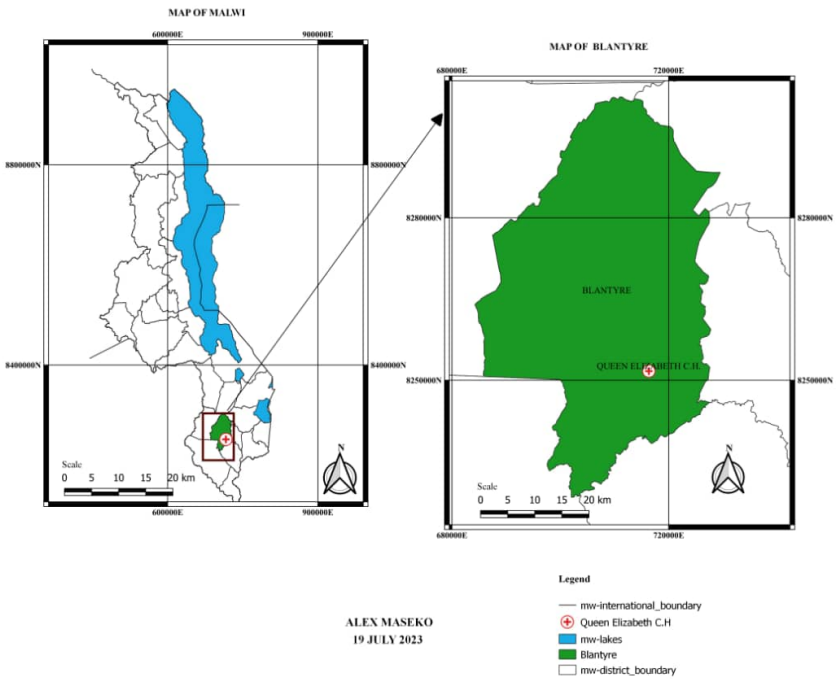

Figure 2: Map of Blantyre showing Queens Central Hospital

6.2 Research Design

- This study used quantitative design where secondary data was collected and used to analyse seasonal patterns, trends and forecast NCDs cases at QECH.

  6.3 Research Framework

- This study included all adults recorded to have NCDs (2010-2022).Data was be collected at QECH in the Health Management Information Systems.

6.5 Data Analysis and Statistical model

- Data Analysis was done using R software with statistical tests done at 5% level of significance.
- Augumented Dickey-Fuller tests for stationarity,
- AIC(akaike information creteria) and BSIC(Bayesian schawatz infomation criterion)
- Autocorrelation for testing if the error terms are linearly correlated.

ARIMA model

- Autoregressive Integrated Moving Average (ARIMA) is an analytical method that formulates time series values as linear functions of past values and random errors (Box, Jenkins, Reinsel, & Ljung, 2015).
- Seasonal ARIMA have p,d,q and P,D,Q are the nonseasonal and seasonal components respectively
- Therefore the ARIMA model combined is;

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + ... + \phi_p y_{t-p} + \epsilon_t - \theta_1 \epsilon_{t-1} - ... - \theta_{t-q} \quad (1)$$

- The ARIMA Model assumes that: error terms are white noise, no outliers, residuals have zero mean and equal variance and residuals are independent.
- The Box-Jenkins approach categorizes into: Model Identification, Parameter estimation and Model diagnosis.(Box et al., 2015)

7.0 RESULTS

Table 1: Descriptive Statistics

Statistic	NCDs Cases
Min.	91.0
1st Qu.	151.0
Median	210.0
Mean	224.1
3rd Qu.	283.5
Max.	447.0

- The minimum recorded number of NCDs cases at QECH was 91, with the highest recorded number of NCDs cases at 447.
- The first quartile (25th percentile) was 151 and the third quartile (75th percentile) was 283.5.
- The mean and median number of recorded NCDs cases were 224.1 and 210 respectively.

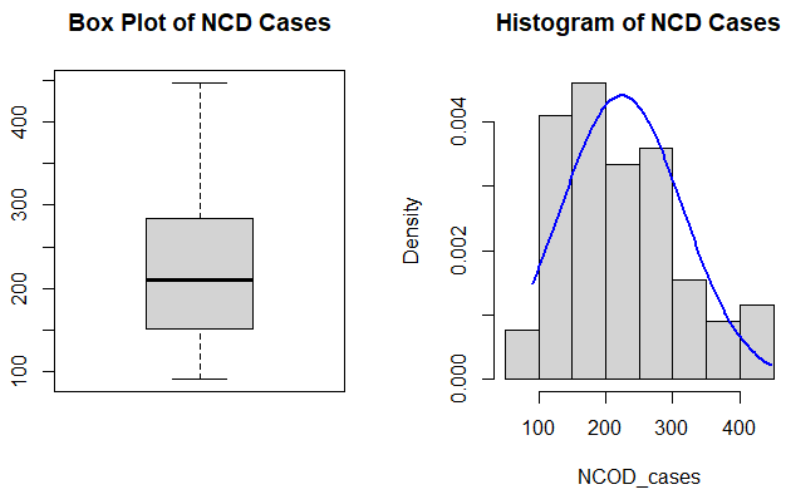

Figure 3: Distribution of NCDs

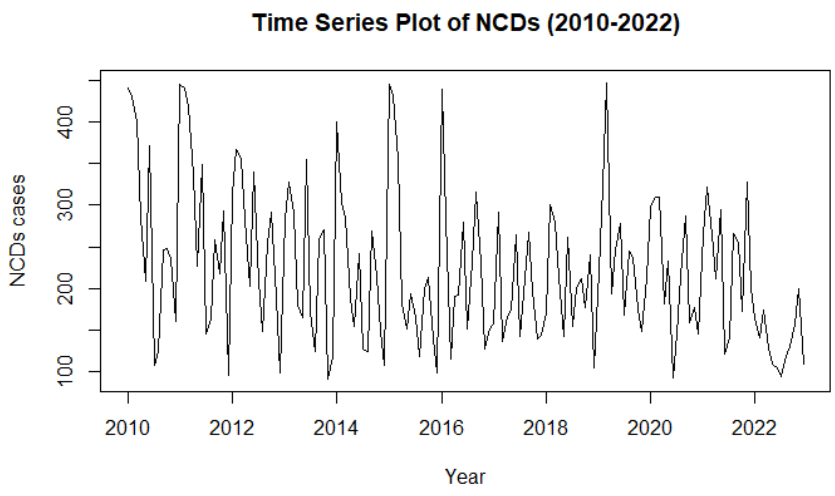

Figure 4: Time Series Plot of NCDs(2010-2022)

- Both the histogram and the Boxplot show that the NCDs cases are skewed to the right, meaning that more NCDs cases are less than the average value.
- The time series plot shows repetitive patterns from 2010 to 2022, but the prevalence of NCDs is decreasing.

7.1 Trend Analysis

- The Time Series data of NCD cases was decomposed to identify the trend component, seasonal component, and a trend analysis was done to find the trend of NCDs.

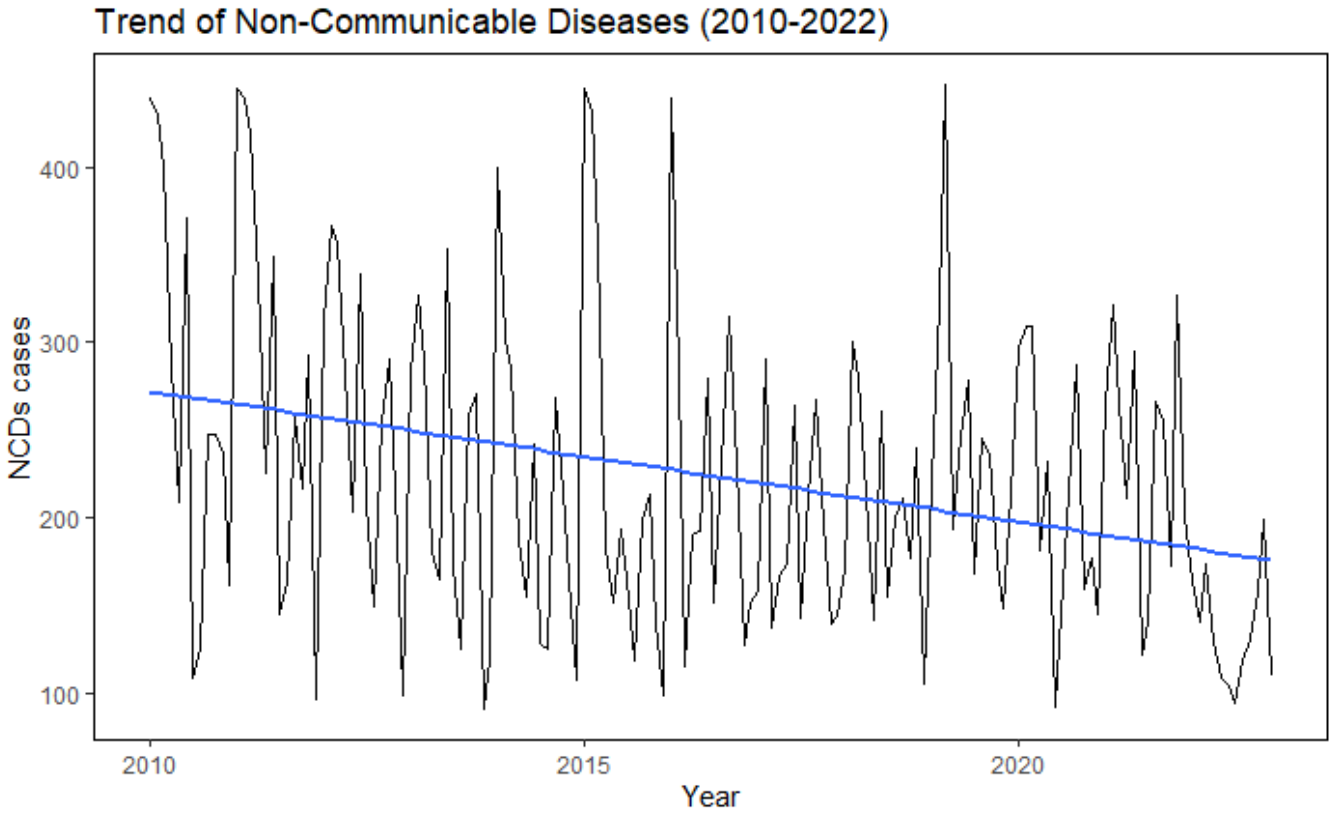

Figure 5: Linear Trend of NCDs

- From figure above, the trend of the NCDs cases is shown to be decreasing, this means that the number of NCDs cases is decreasing from the year 2010 to 2022.

Table 2: Trend Analysis

Coefficients	**Intercept**	**t**
Statistic	272.9733	-0.6228
Std error	13.8651	0.1532
t-statistic	19.688	-4.065
$Pr(>\|t\|)$	$< 2e-16$	$7.63-e05$

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R-squared: 0.4114, Adjusted R-squared: 0.3997 F-statistic:
16.53 on 1 and 154 DF, p-value: 7.63e-05
```

- The intercept of 272.9733 suggests that when time is zero, the predicted value of NCDs cases is 273. The slope of -0.6228 indicates that, on average, for each unit increase in time, the predicted value of NCDs decreases by about 1.
- The p-values for both the intercept and the time variable are very low (less than 0.05), suggesting that both are statistically significant, hence including both intercept and time variable was appropriate.
- The R-squared value of 0.4114 indicates that the model explains approximately 41.14% of the variance in the response variable.
- The F-statistic and its associated p-value suggest that the overall model is statistically significant.

7.2 Seasonality Analysis

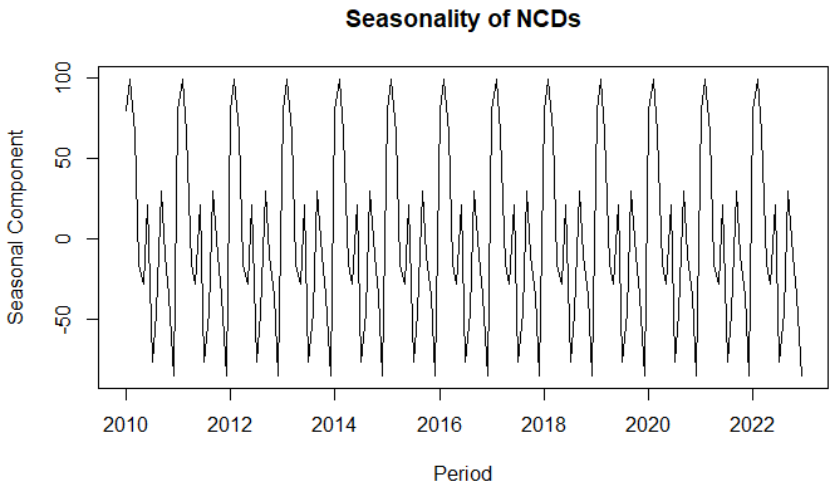

Figure 6: Seasonality of NCDs

- The figure shows that there is a repetitive pattern of peaks and troughs,suggesting the presence of seasonality.

7.3 Model Building

- The data was first tested for presence of unit root (nonstationarity)

Table 3: ADF test for unit root

ADF test	
ADF statistic	-7.1052
P-value	0.01

- From table , the p-value (0.01) is smaller than significance level (0.05), hence the conclude that the data is stationary by rejecting the null hypothesis of nonstationary or unit root. .

- Since the data exhibited seasonality, the data was seasonally differenced to remove the seasonality

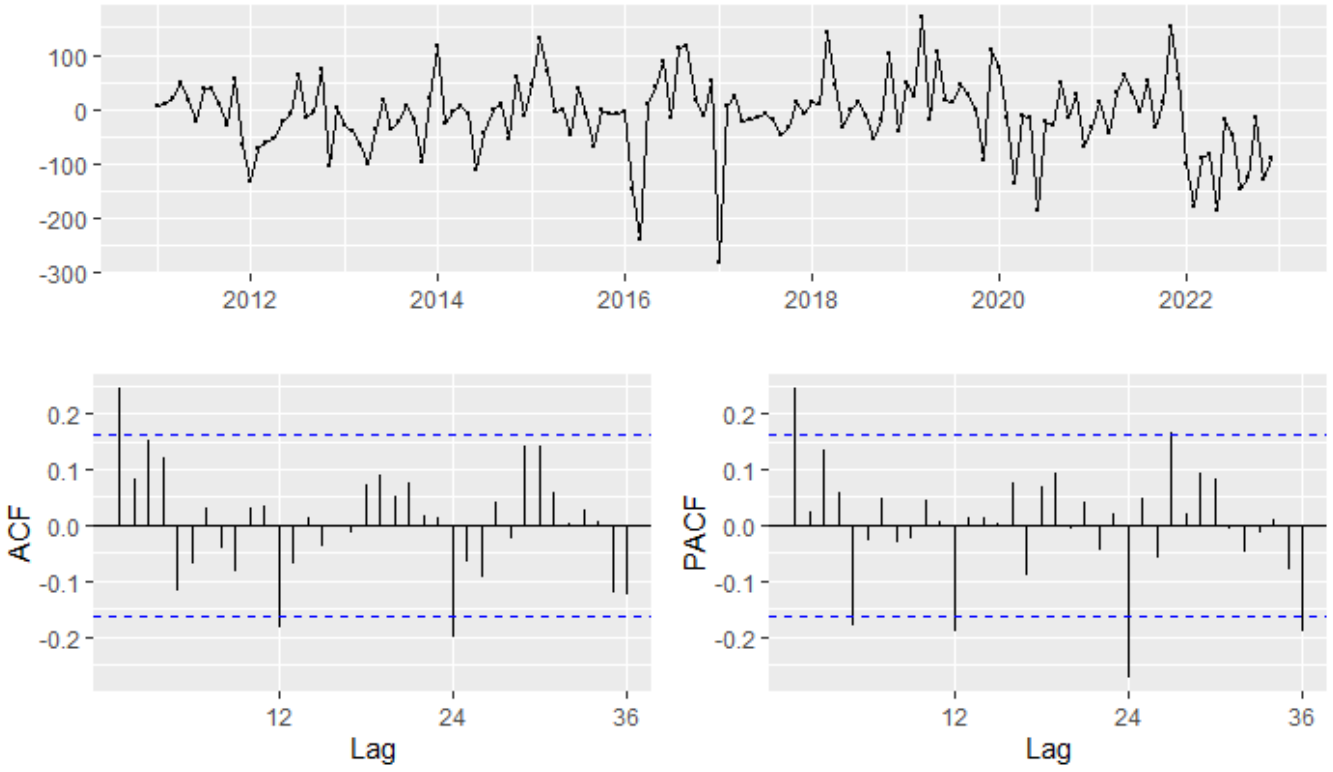

Figure 7: Seasonally Differenced time series of NCDs

- From inspecting the ACF and PACF of figure 6, the orders of the model were chosen

Table 4: ARIMA (0,0,1)(0,1,2)[12] with drift

Coefficients	**Value**	**Std Error**	**z value**	$Pr(>\|z\|)$
ma1	0.2567	0.080809	3.1768	0.001489
sma1	-0.3939	0.086529	-4.5522	$5.31e-06$
sma2	-0.2711	0.086148	-3.1474	0.001647
drift	-0.6717	0.24069	-2.7876	0.005309

Table 5: ARIMA (1,0,1)(0,1,1)[12]

Coefficients	**Value**	**Std Error**	**z value**	$Pr(>\|z\|)$
ar1	0.61006	0.22660	2.6923	0.007097
ma1	-0.35702	0.26823	-1.3310	0.183185
sma1	-0.49068	0.10173	-4.8232	1.412e-06

```
Accuracy measures for ARIMA (0,0,1)(0,1,2)[12] with drift
RMSE        MAPE         R-Sq.       Adj.R-Sq
58.92199    21.78797     0.5722796   0.5609493

sigma^2 =  3869:      log  likelihood = -800.08
AIC     =   1610.17    AICc =1610.6    BIC=1625.02
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Accuracy measures for ARIMA (1,0,1)(0,1,1)[12]
RMSE       MAPE        R-Sq.       Adj. R-Sq.
62.16529   22.67354   0.5238967   0.5112837

sigma^2 = 4276:  log likelihood = -806.49
AIC     =1620.98   AICc=1621.26   BIC=1632.85
```

- ARIMA (0,0,1)(0,1,2)[12] with drift was the model chosen for prediction since it has the lowest AIC, AICc, BIC, RMSE and MAPE
- The model also has a higher R-Squared value hence it expplains more of variations in the response variable.

7.4 Model Diagnostics

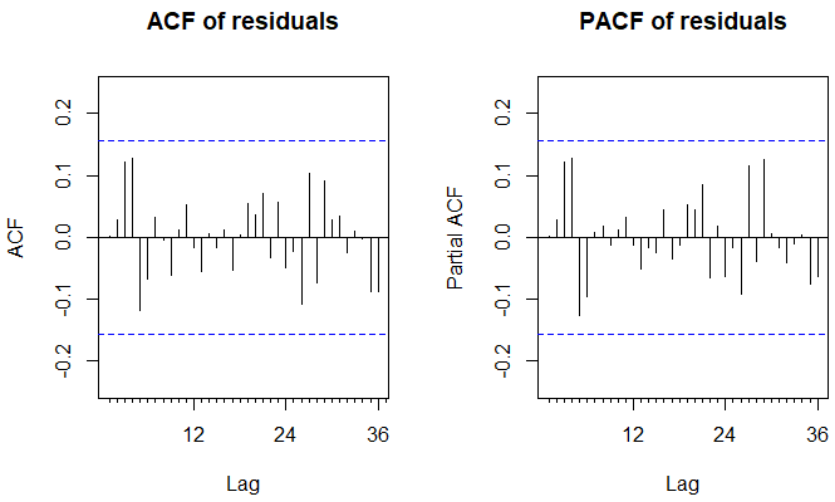

Figure 8: ACF and PACF of Model Residuals

- The figure shows that all ACF and PACF values are within the confidence intervals, it suggests that the residuals are independent and do not exhibit significant autocorrelation. .

- The residuals were tested for Gaussian White Noise

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 Ljung-Box test

data: Residuals from ARIMA(0,0,1)(0,1,2)[12] with drift
Q* = 13.571          p-value = 0.8873
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- The p-value (0.8873) is greater than 0.05 significant level meaning we cannot reject the null hypothesis, in other words accepting the null hypothesis means that the residual variable is white noise

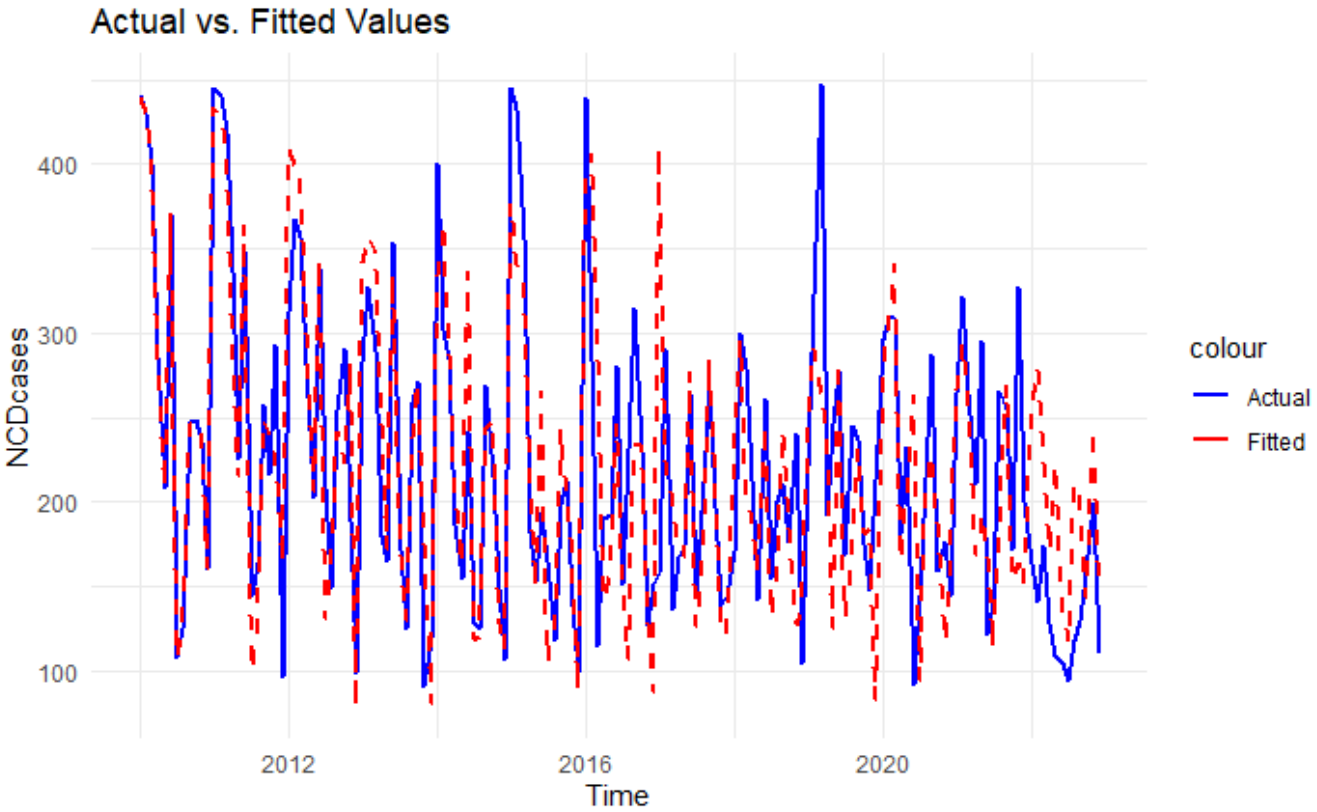

Figure 9: Actual V Fitted Data

- The primary purpose is to visually assess how well the model fits the observed data.
- In this case,it can be shown that the fitted values are close to the actual data,indicating the goodness of fit of the model

7.5 Forecasting

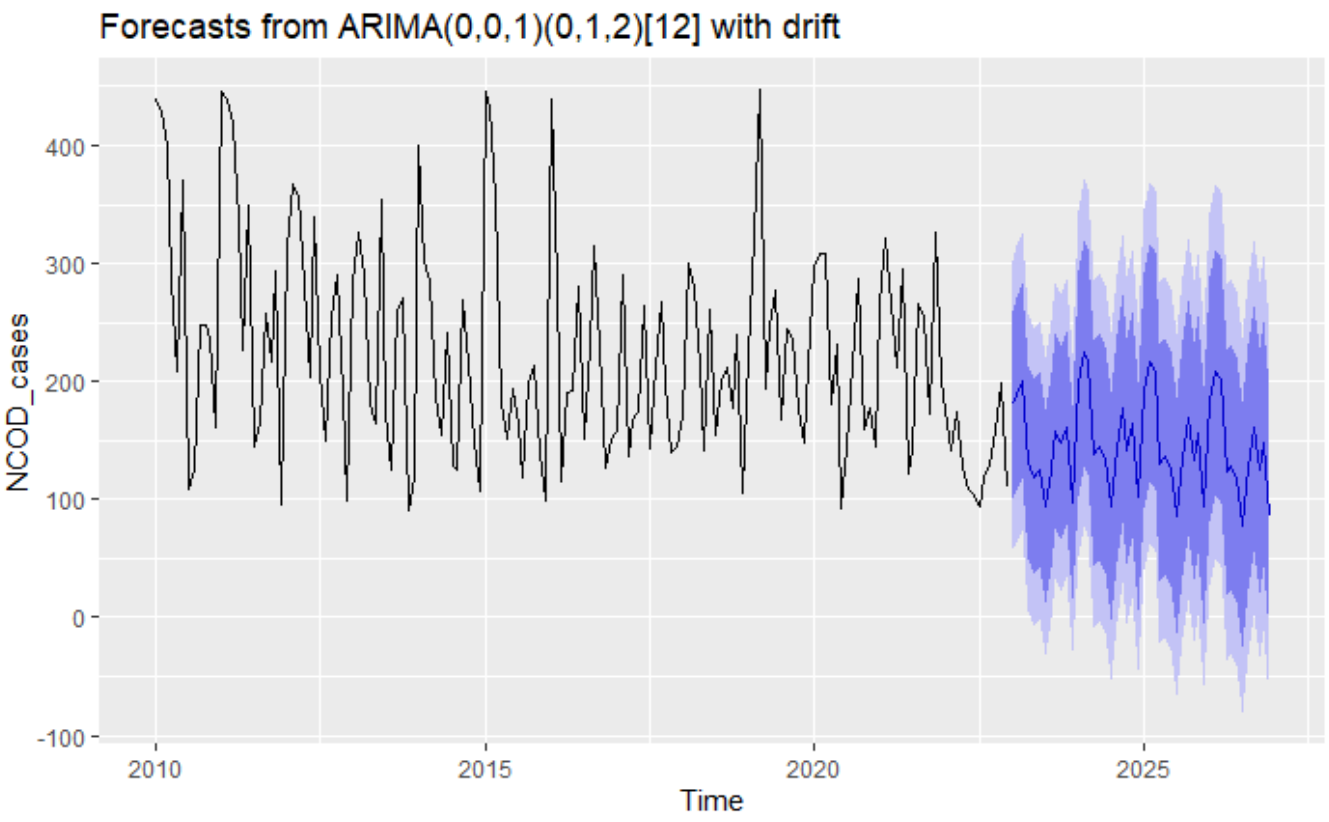

Figure 10: Forecasts from chosen model

- The forecasts shows a decreasing forecasting trend, but periodic patterns of increase or decrease have to be taken into account.

8.0 Conclusions and Recommendations

8.1 Conclusion

- The trend analysis indicated a negative slope meaning that there is a decrease in the recorded cases of NCDs at Queen Elizabeth Central Hospital.
- There is presence of seasonality in NCD cases at QECH.
- The forecast has shown that there will continue to be a decrease in NCDs cases.

8.2 Recommendations & Area of further study

- Identify opportunities for further research, especially in understanding the root causes of NCDs.
- Investigate the impact of socioeconomic factors, environmental influences, and emerging health risks on NCDs.
- There is presence of seasonality in NCD cases at QECH. Therefore, the hospital has to investigate and understand the factors that may be influencing the observed seasonality.
- Consider external factors such as changes in demographics, lifestyle, healthcare policies.

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